Algorithms

Lecture # 28

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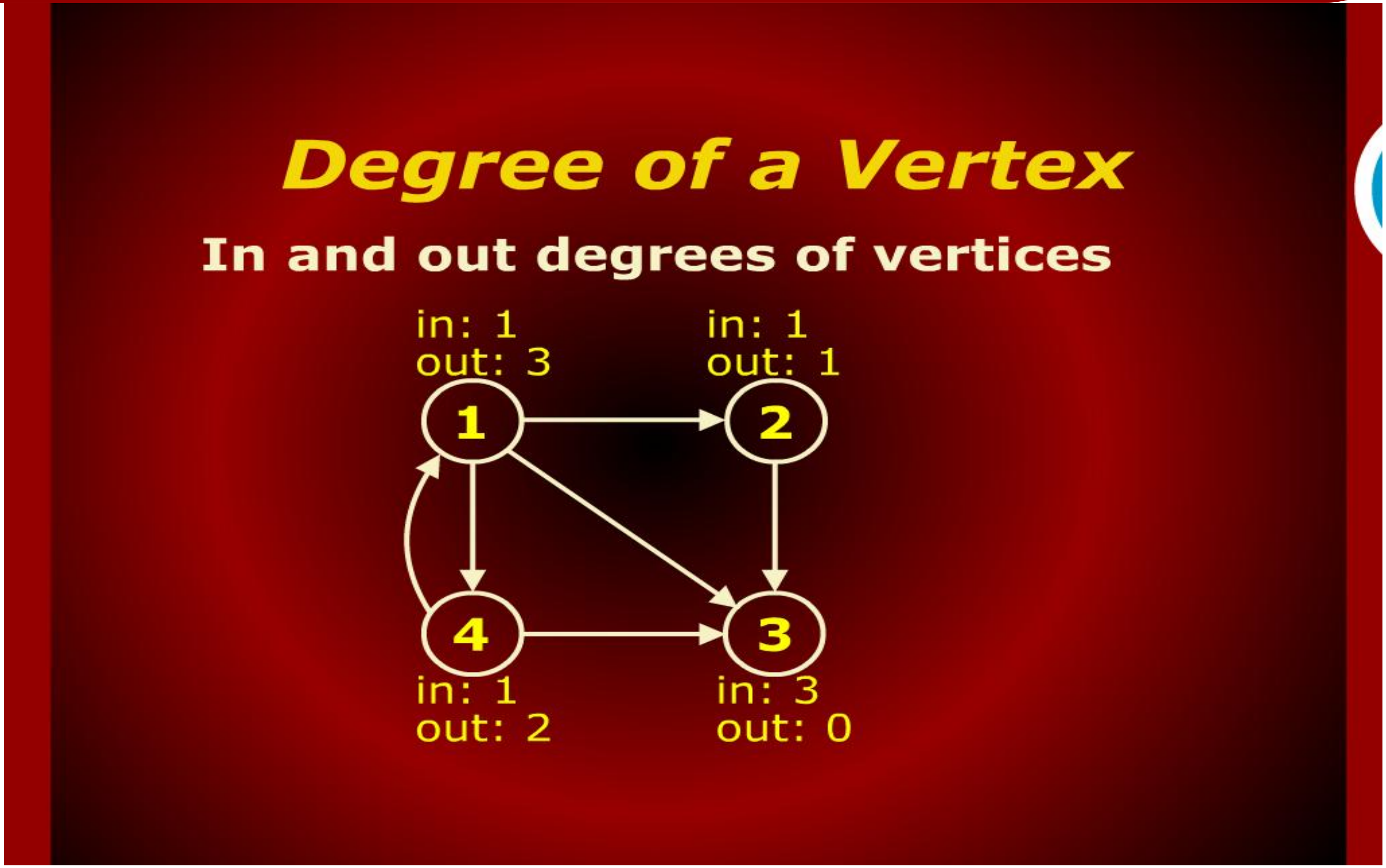
Degree of a Vertex

Degree of a Vertex

- In a digraph, the number of edges coming out of a vertex is called the out-degree of that vertex.
- Number of edges coming in is the in-degree.
- In an undirected graph, we just talk of degree of a vertex. It is the number of edges incident on the vertex.

Degree of a Vertex



Observations about Graphs

Observations about Graphs

For a digraph G = (V, E),

$$\sum_{v \in V} \text{in-degree}(v) \sum_{v \in V} \text{out-degree}(v) = |E|$$

where |E| means the cardinality of the set E, i.e., the number of edges

Observations about Graphs

Observations about Graphs

For an undirected graph G=(V, E),

$$\sum_{v \in V} \text{degree}(v) = 2|E|$$

where |E| means the cardinality of the set E, i.e., the number of edges.

Paths and cycles

Paths and Cycles

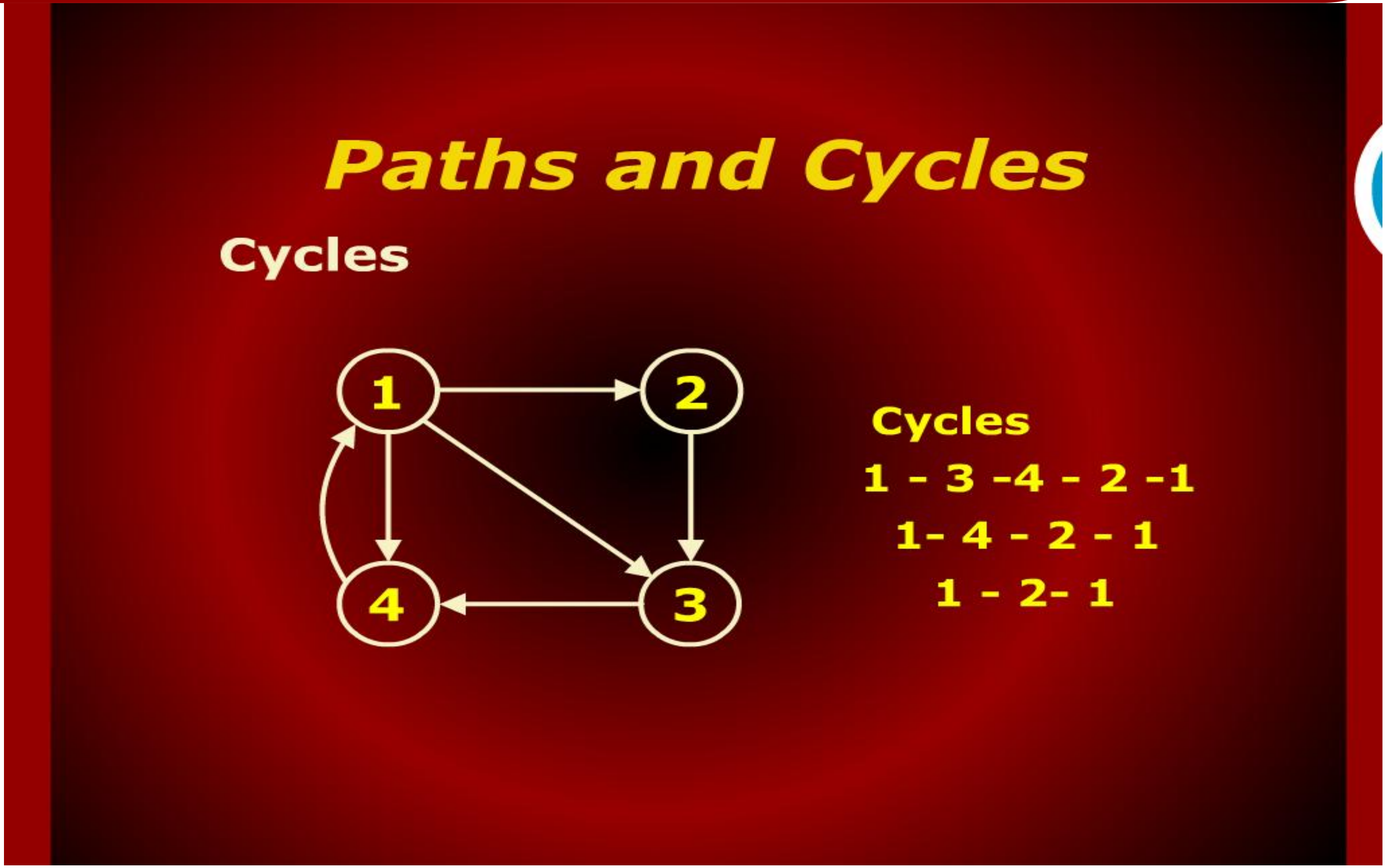
- A path in a directed graphs is a sequence of vertices $\langle v_0, v1, . . , v_k \rangle$ such that $(v_i\text{-}1, v_i)$ is an edge for $i = 1, 2, . . . , k$.
- The length of the paths is the number of edges, k.
- A vertex w is reachable from vertex u is there is a path from u to w.
- A path is simple if all vertices (except possibly the fist and last) are distinct.

Paths and cycles

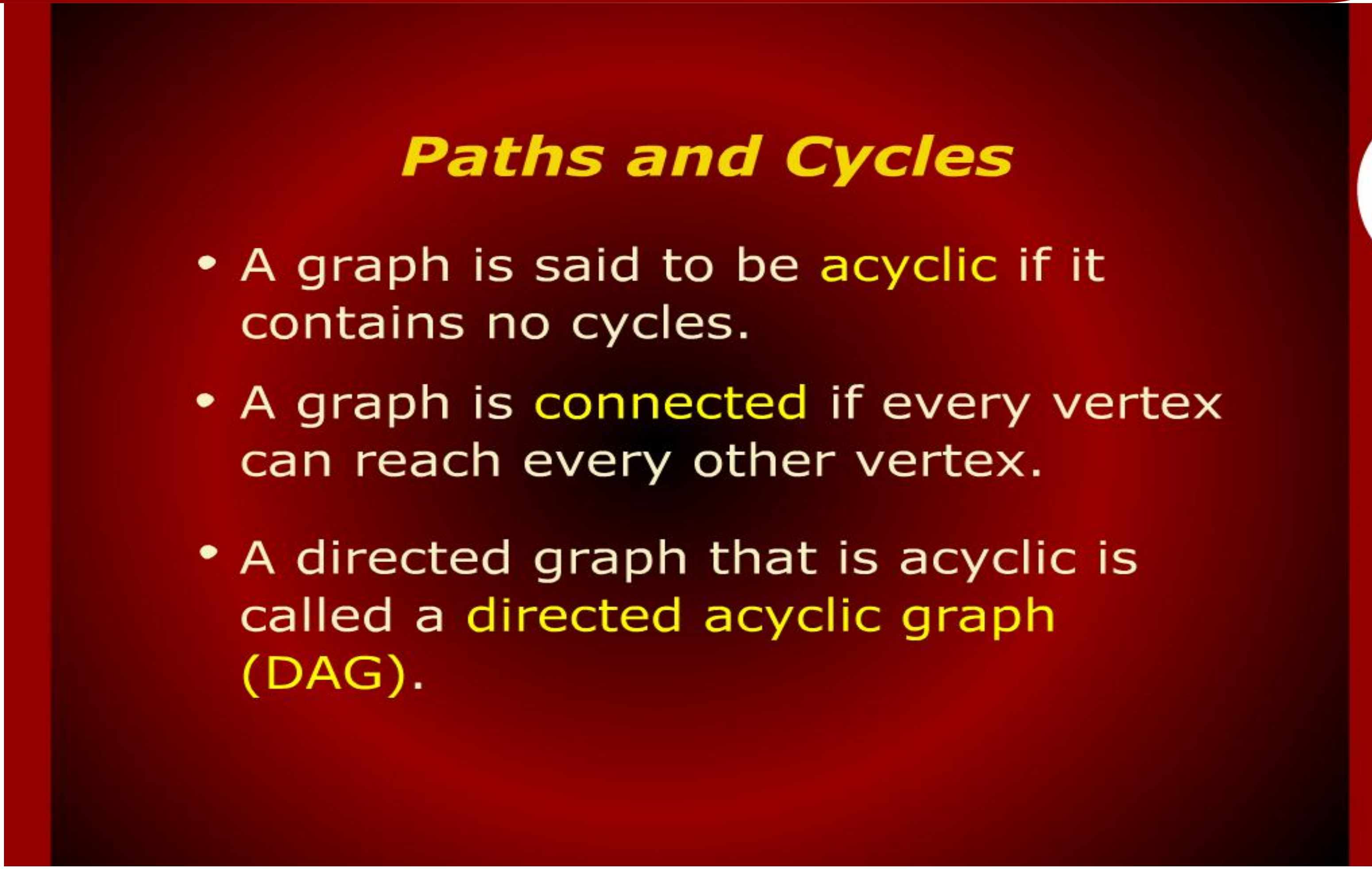
Paths and Cycles

- A cycle in a digraph is a path containing at least one edge and for which $v_0 = v_k$.
- A Hamiltonian cycle is a cycle that visits every vertex in a graph exactly once.
- A Eulerian cycle is a cycle that visits every edge of the graph exactly once.
- There are also "path" versions in which you do not need return to the starting vertex.

Paths and cycles



Paths and cycles



Graph Representations

Graph Representations

- There are two ways of representing graphs:
  - adjacency matrix
  - adjacency list
- Let G = (V, E) be a digraph with n = |V| and let e = |E|.
- We will assume that the vertices of G are indexed {1, 2, . . . , n}.

Graph Representations

Graph Representations

Adjacency Matrix

- An $n \times n$ matrix defined for $1 \le v,w \le n$.

$$A[v,w] = \begin{cases} 1 & \text{if } (v,w) \in E \\ 0 & \text{otherwise} \end{cases}$$

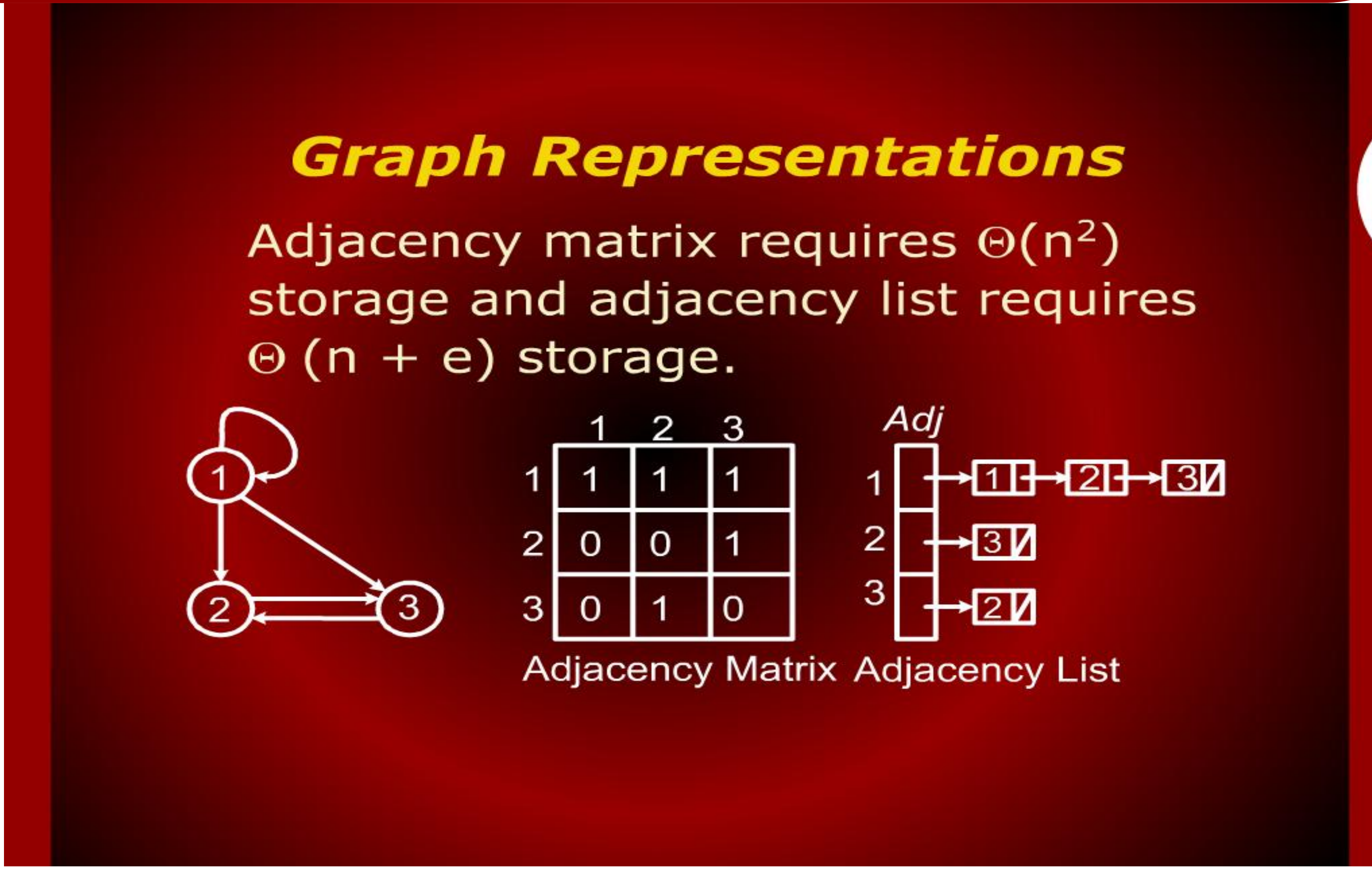
Graph Representations

Graph Representations

Adjacency List

- An array Adj[1..n] of pointers where for $1 \leq v \leq n$, Adj[v] points to a linked list containing the vertices which are adjacent to v

Graph Representations



Shortest Path

Shortest Path

To motivate our first algorithm on graphs, consider the following problem.

- We are given an undirected graph $G = (V, E)$ and a source vertex $s \in V$.
- The length of a path in a graph is the number of edges on the path.
- We would like to find the shortest path from s to each other vertex in the tree.

Shortest Path

Shortest Path

The final result will be represented in the following way.

- For each vertex $v \in V$, we will store $d[v]$ which is the distance (length of the shortest path) from s to v. Note that $d[s] = 0$.
- We will also store a predecessor (or parent) pointer $\pi[v]$ which is the first vertex along the shortest path if we walk from v backwards to s. We will set $\pi[s] = Nil$.

Shortest Path

Shortest Path

- There is a simple brute-force strategy for computing shortest paths.
- We could simply start enumerating all simple paths starting at s, and keep track of the shortest path arriving at each vertex.
- However, there can be as many as n! simple paths in a graph.
- Clearly this is not feasible.

Design and Analysis

Of

Algorithms

GRAPHS